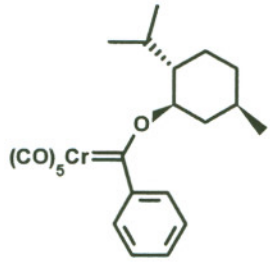
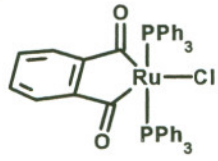
**Inorganic Cume, December 5, 2003**

1. Give the d-electron count (use oxidation state method) of the following compounds. Give the formal oxidation state of the metal.

A.




B.




C.

$[PtCl_3(C_2H_4)]^-$

D.

$[Ir(CO)(PPh_3)_2(Cl)(NO)]^+$

E.

$[HFe(CO)_4]^-$

F.

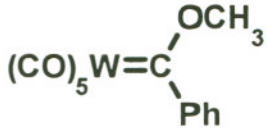
$(diphos)Pt(OMe)_2$

2. For each of the following metal and ligand combinations, formulate the simplest neutral compound that conforms to the 18-electron rule. Draw a reasonable structure for each compound.

   A. Ni, CO
   B. Fe, CO, COT = cyclooctatetraene
   C. Re, CO, H

3. Give one example of each of the following processes.
   a. Associative process
   b. Dissociative process
   c. Oxidative addition
   d. β-hydride elimination

4. Show how the following compound can be prepared from a tungsten hexacarbonyl.




5. Explain the relative values for the $\Delta G^{\ddagger}_{rot}$, for the rotation around the metal-carbon bond for $Cp(CO)_2(PPh_3)W{=}CH_2^+$ having $\Delta G^{\ddagger}_{rot}$ = 8.3 kcal/mol and $Cp(CO)_2(PEt_3)W{=}CH_2^+$ with $\Delta G^{\ddagger}_{rot}$ = 9.0 kcal/mol.